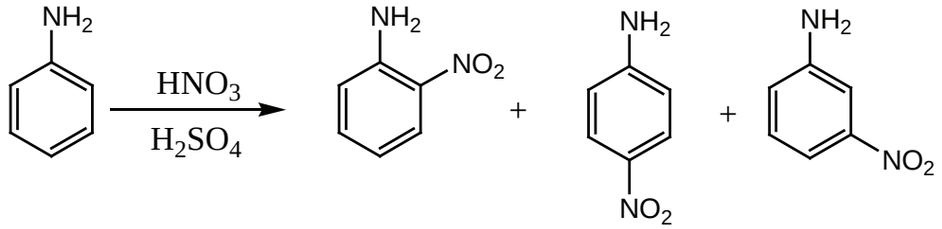
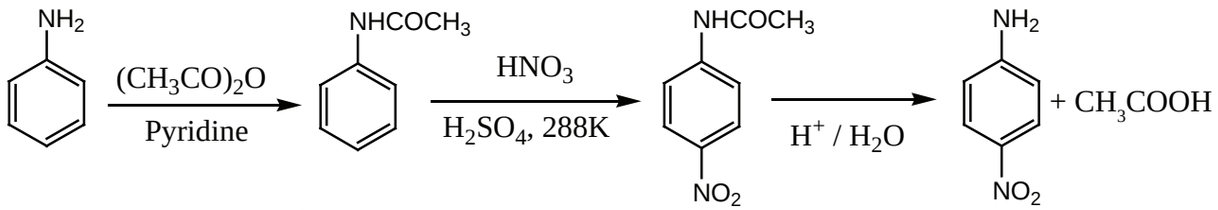
ii) Nitration

Direct nitration of aniline gives o and p – nitro aniline along with dark coloured 'tars' due to oxidation. Moreover in a strong acid medium aniline is protonated to form anilinium ion which is m – directing and hence m – nitro aniline is also formed.



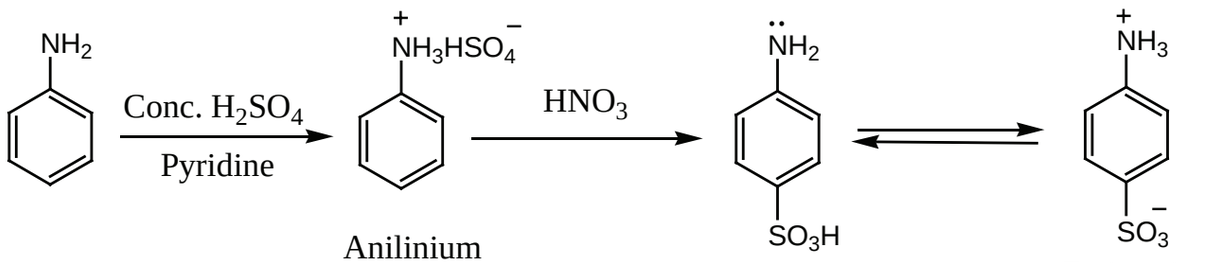
To get para product, the - NH_2 group is protected by acetylation with acetic anhydride. Then, the nitrated product is hydrolysed to form the product.



Acetanilide — P-nitroacetanilide — p-nitro aniline

iii) Sulphonation

Aniline reacts with Conc. H_2SO_4 to form anilinium hydrogen sulphate which on heating with H_2SO_4 at 453 – 473K gives p- aminobenzene sulphonic acid, commonly known as sulphanilic acid, as the major product.



Anilinium hydrogensulphate — Sulphanilic acid — Zwitter ion

iv) Aniline

It does not under go Friedel – Crafts reaction (alkylation and acetylation) we know aniline is basic in nature and it donates its lone pair to the lewis acid $AlCl_3$ to form an adduct which inhibits further the electrophilic substitution reaction.

13.3 DIAZONIUM SALTS

13.3.1 Introduction

We have just learnt that aromatic amines on treatment with $(NaNO_2+HCl)$ gives diazonium salts. They are stable only for a short time and hence are used immediately after preparation.

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